

ECG Heartbeat Classification Model report

Project Overview:

Electrocardiograms (ECGs) capture the electrical activity of the heart, providing vital information for diagnosing normal rhythms and abnormalities such as arrhythmias or myocardial infarction. This project aims to develop machine learning and deep learning models to classify ECG signals effectively. Specifically, we evaluate models including Convolutional Neural Networks (CNNs) and XGBoost to identify the most accurate approach for detecting arrhythmia patterns in ECG data.

Github Repository : https://github.com/Saikot1997/Hearbit_Classification_Datascientist

Dataset Source:

The dataset utilized in this project is the ECG Heartbeat Categorization Dataset, available on [Kaggle](#).

This dataset comprises two collections of heartbeat signals derived from the MIT-BIH Arrhythmia Dataset and the PTB Diagnostic ECG Database.

The dataset includes the following CSV files:

- **MIT-BIH Dataset:**
 - [mitbih_train.csv](#): Training data.
 - [mitbih_test.csv](#): Testing data.
- **PTB Dataset:**
 - [ptbdb_abnormal.csv](#): Abnormal ECG samples.
 - [ptbdb_normal.csv](#): Normal ECG samples.

Each file contains preprocessed and segmented ECG signals, with each row representing a single heartbeat. The last column in each row indicates the class label.

Dataset Exploration & Visualisation

MIT-BIH Dataset

Number of Samples: 109446

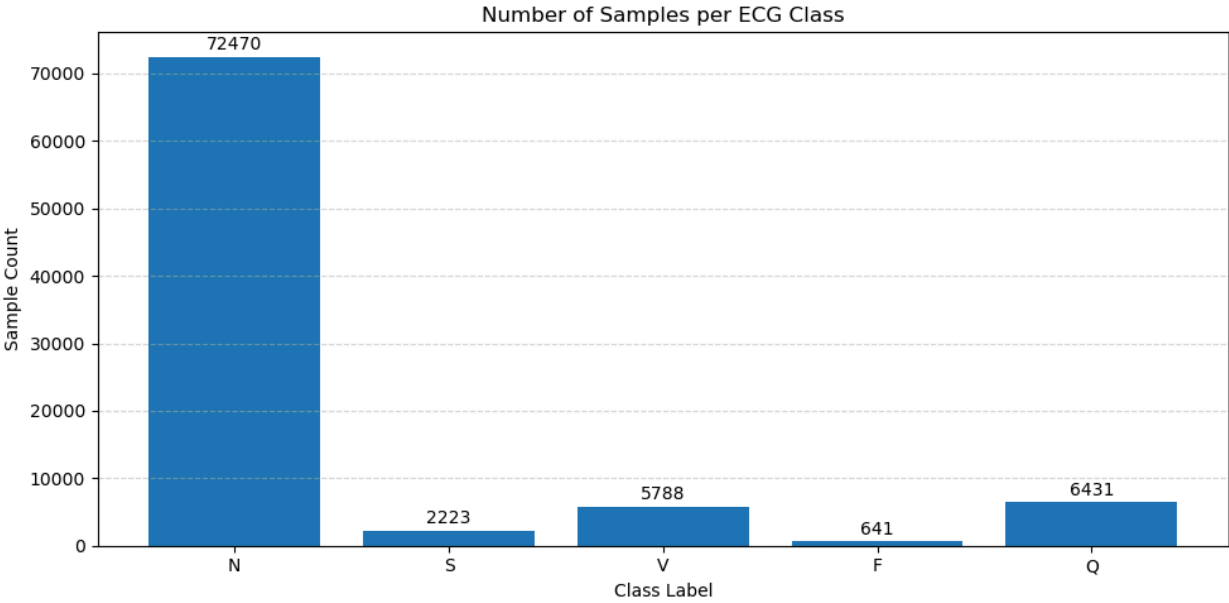
Number of Categories: 5

Sampling Frequency: 125Hz

Data Source: Physionet's MIT-BIH Arrhythmia Dataset

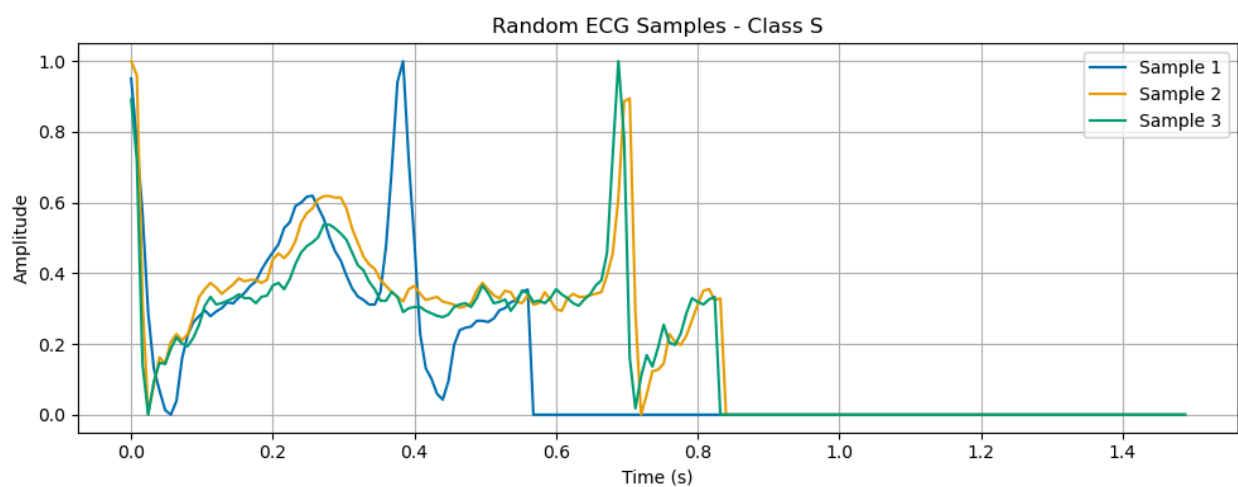
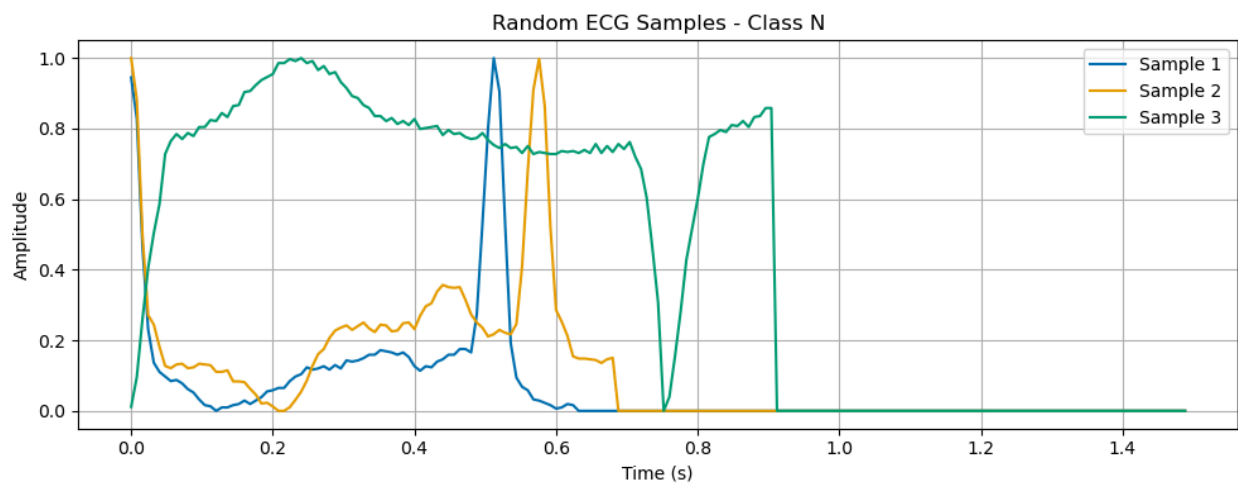
Classes: ['N': 0, 'S': 1, 'V': 2, 'F': 3, 'Q': 4]

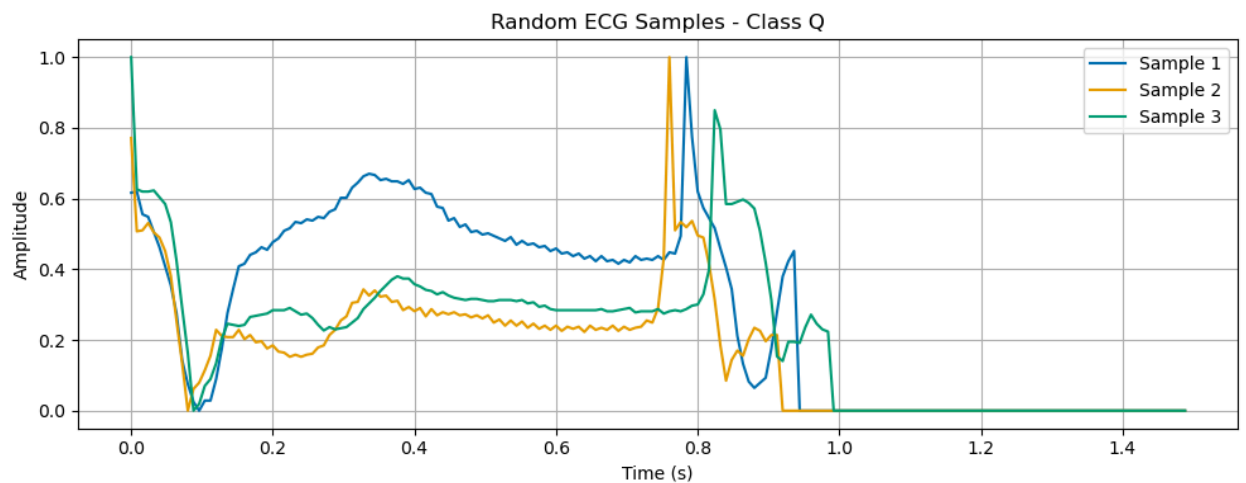
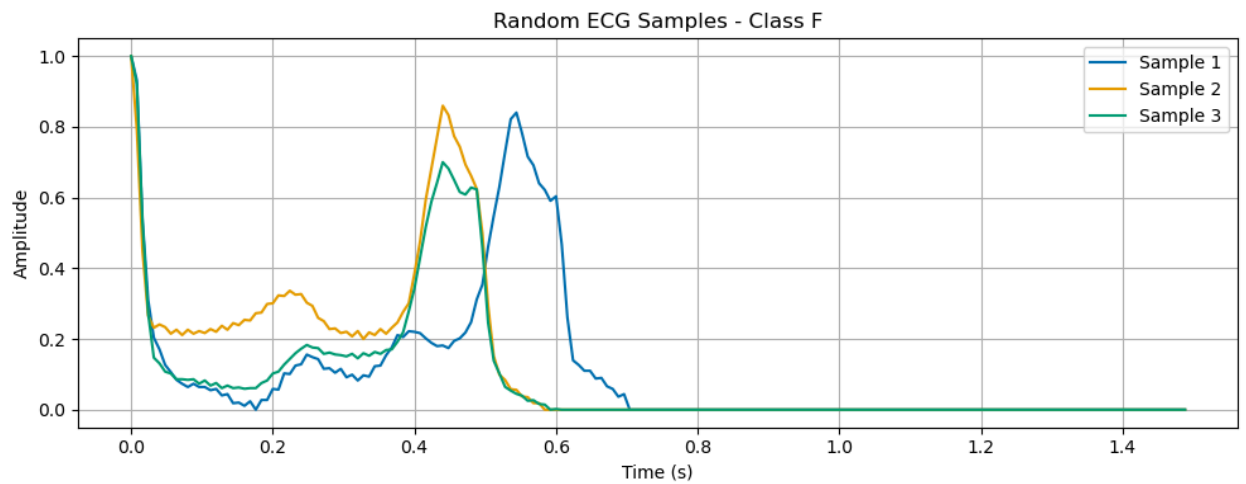
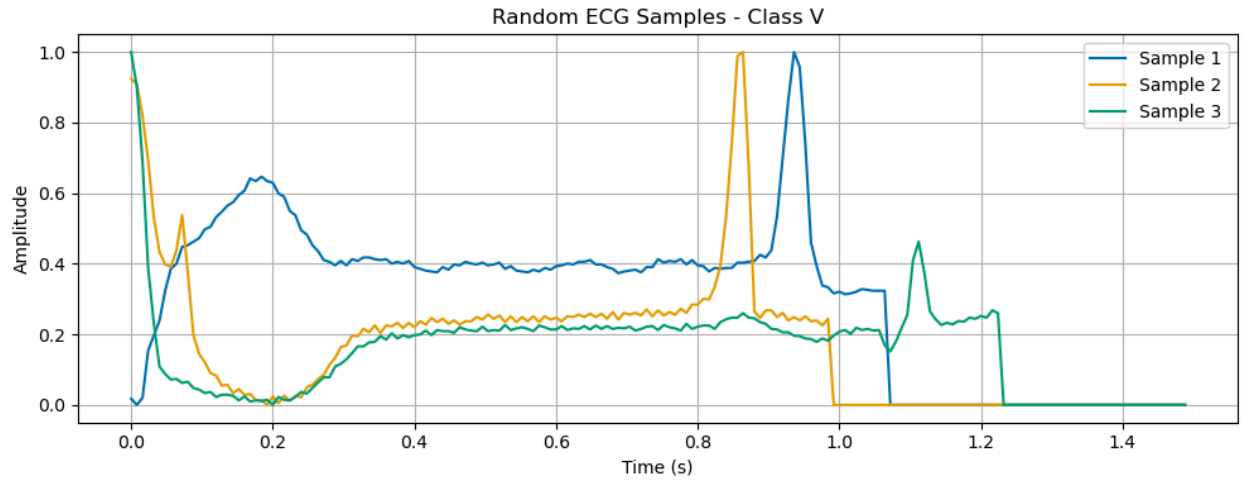
Category	Code	Meaning	Indication
N	0	Normal beat	Healthy
S	1	Supraventricular ectopic beat	Pathological
V	2	Ventricular ectopic beat	Pathological
F	3	Fusion beat	Pathological
Q	4	Unknown beat	Potentially pathological



The number of samples in the Normal beat category is extremely high, while the Supraventricular Ectopic Beat and Fusion Beat categories have relatively few samples. This imbalance may negatively impact the classification accuracy for these underrepresented classes.

- **Random Signal Visualization**





- **Resampling**

Target: around 20000 entries for each class, under-sampling for N, over-sampling for other classes.

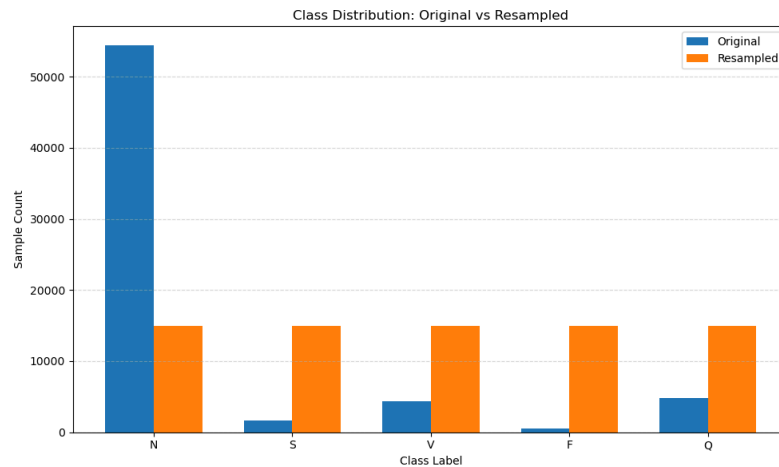
Under-sampling:

RandomUnderSampler

Over-sampling:

SMOTE-Tomek to 20000 each class

GAN to 15000 each class due to separate val set



PTB Diagnostic ECG Dataset

The PTB Diagnostic ECG dataset comprises two CSV files:

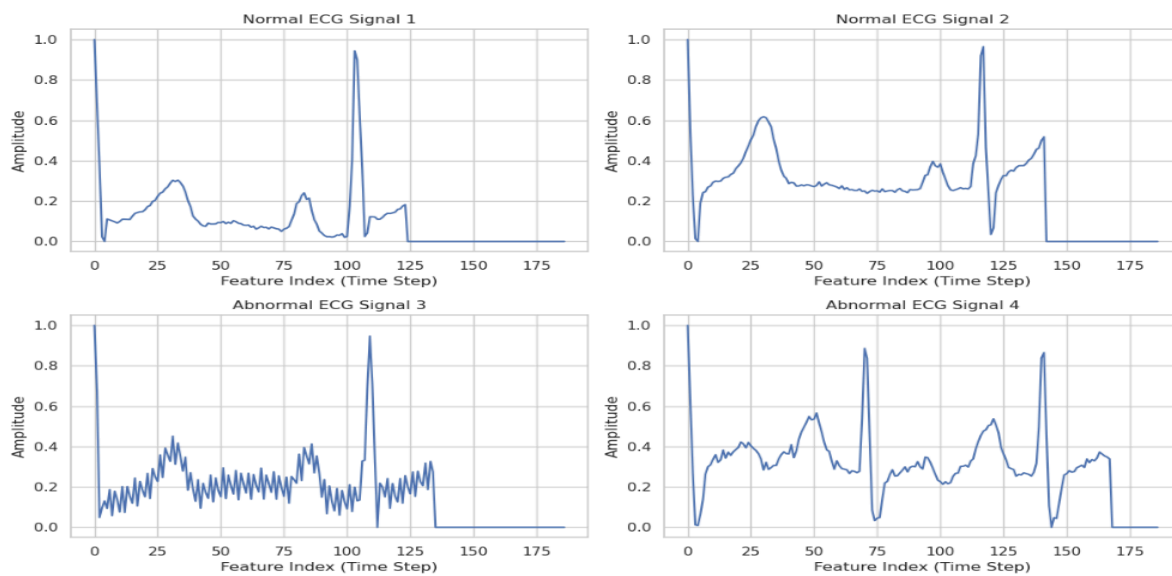
- **ptbdb_normal.csv**: Contains 10,506 rows and 188 columns, representing normal ECG readings.
- **ptbdb_abnormal.csv**: Contains 4,046 rows and 188 columns, representing abnormal ECG readings.

When concatenated using `pd.concat([ptbdb_normal, ptbdb_abnormal], ignore_index=True)`, the resulting combined dataset consists of 14,552 rows and 188 columns. There are no missing values in the dataset, and the unique value count indicates that the target column (the last column) consists of two binary classes: '0' for normal and '1' for abnormal.



Below are visualizations of four randomly selected ECG signal samples from the dataset:

- Two samples labeled as 'normal' (Class 0)
- Two samples labeled as 'abnormal' (Class 1)



Implementation of Machine Learning Models for MITBIH Diagnostic ECG Dataset

XGBoost for comparison

Since this is a medical classification task, recall is prioritized over accuracy, especially for classes representing pathological conditions. In a clinical context, failing to detect a disease (false negative) is generally more critical than raising a false alarm (false positive).

In the test data, the category distribution is still various as following:

N 18117
S 556
V 1448
F 162
Q 1608

In the result we are going to use macro avg recall as our benchmark, for two props:

Missing a diagnosis (false negative) can be fatal, false positives can be checked by further test.

The imbalance data set is common. We need a benchmark not masked by the dominant class.

The macro average recall is chosen as the benchmark.

The RandomizedSearchCV has been used to ensure to find the good result without consuming too much resources.

The gamma and reg_alpha are fixed to prevent overfitting.

Model	Best Parameters	Best Score
XGBoost	subsample: 0.6, reg_alpha: 0.5, n_estimators: 300, min_child_weight: 5, max_depth: 5, learning_rate: 0.1, gamma: 0.8, colsample_bytree: 0.8	0.898

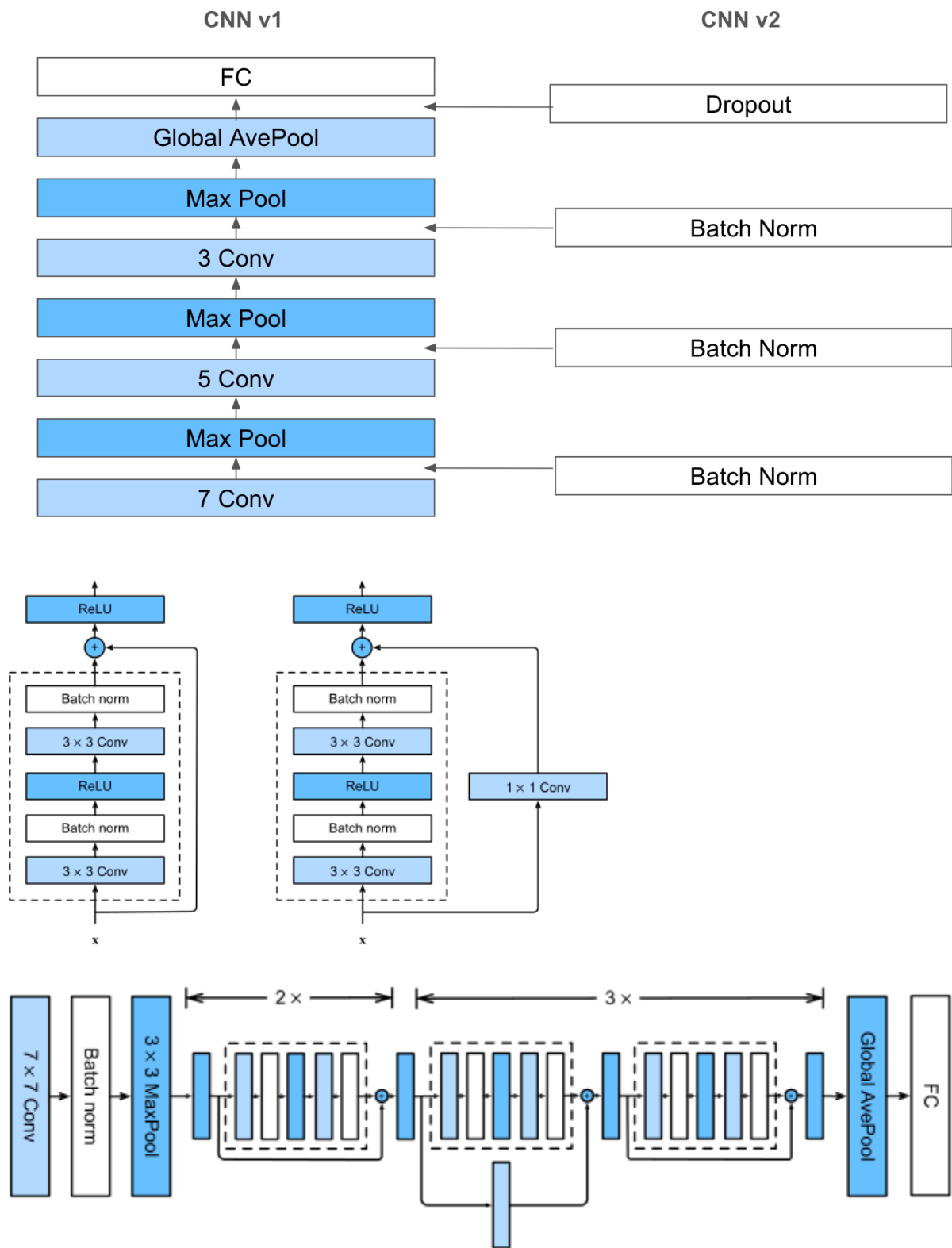
Neural Network Model Training

Three models including

CNN v1, 3 convolutional layers with maxpool

CNN v2, on top of CNN v1, added batch normalization and dropout to prevent overfitting.

ResNet1D, changed from standard ResNet-18 to 1 dimensional convolution, added dropout before final fc as well.



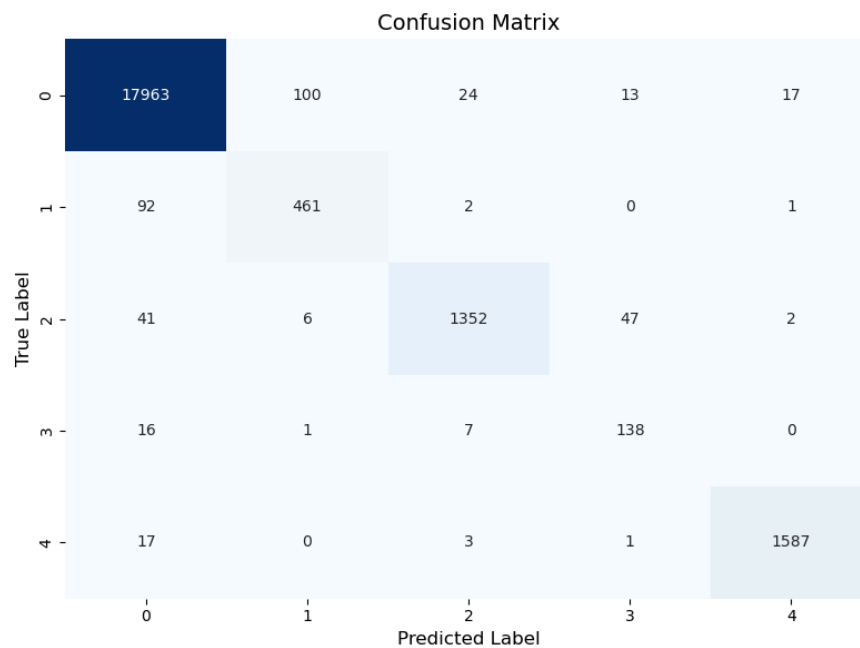
The performance on validation dataset

Model	Macro avg recall	Accuracy
CNN v1	0.828	0.976
CNN v2	0.866	0.981
ResNet	0.934	0.985

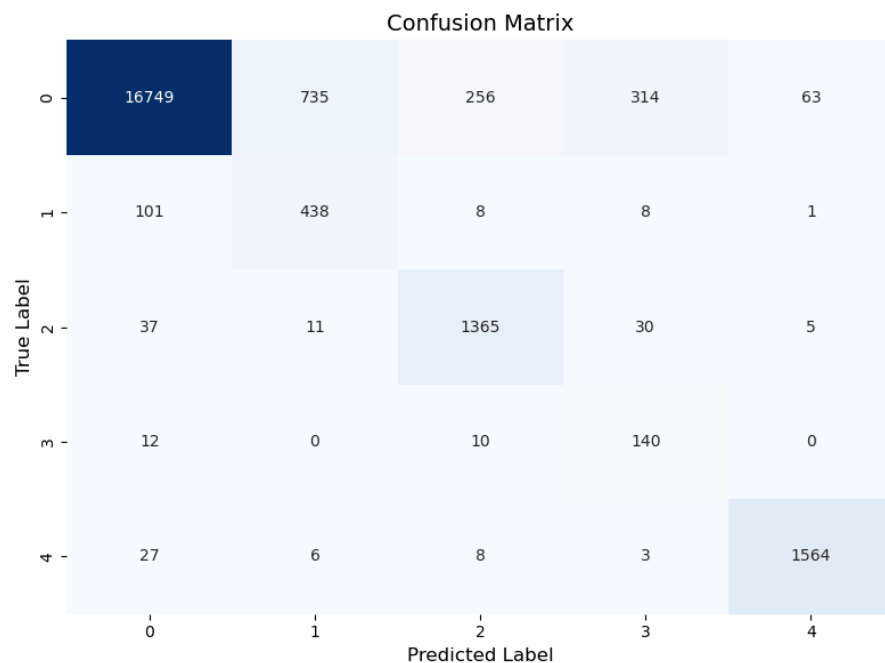
The performance on test dataset

Model	Macro avg recall	Accuracy
ResNet	0.919	0.982
XGBoost	0.898	0.925

ResNet Confusion Matrix



XGBoost Confusion Matrix



Implementation of Machine Learning Models for PTB Diagnostic ECG Dataset

Before implementing deep learning models, multiple machine learning models were applied to the dataset. The models tested include:

1. Random Forest Pipeline
2. SVM Pipeline
3. Logistic Regression Pipeline
4. XGBoost Pipeline
5. K-Nearest Neighbors Pipeline
6. Gradient Boosting Pipeline
7. AdaBoost Pipeline
8. Decision Tree Pipeline
9. CatBoost Pipeline

StandardScaler was used for feature scaling, and all models were implemented using the **Pipeline** structure.

GridSearchCV was applied for hyperparameter tuning using the following parameter grids:

Model	Hyperparameters
Random Forest	n_estimators : [50, 100, 200], max_depth : [5, 10, 15], min_samples_split : [2, 5, 10]
SVM	C : [0.1, 1, 10], kernel : ['linear', 'rbf'], gamma : ['scale', 'auto']

Logistic Regression	C: [0.1, 1, 10], solver: ['liblinear', 'saga']
XGBoost	n_estimators: [50, 100, 200], learning_rate: [0.01, 0.1, 0.2], max_depth: [3, 5, 7]
KNN	n_neighbors: [3, 5, 7, 10], weights: ['uniform', 'distance']
Gradient Boosting	n_estimators: [50, 100, 200], learning_rate: [0.01, 0.1, 0.2], max_depth: [3, 5, 7]
AdaBoost	n_estimators: [50, 100, 200], learning_rate: [0.01, 0.1, 1]
Decision Tree	max_depth: [5, 10, 15], min_samples_split: [2, 5, 10]
CatBoost	iterations: [50, 100, 200], learning_rate: [0.01, 0.1], depth: [3, 5, 7]

Model Training and Tuning Results

Model	Best Parameters	Best Score
Random Forest	max_depth: 15, min_samples_split: 2, n_estimators: 200	0.9659
SVM	C: 10, gamma: 'scale', kernel: 'rbf'	0.9531
Logistic Regression	C: 1, solver: 'liblinear'	0.8236
XGBoost	learning_rate: 0.2, max_depth: 7, n_estimators: 200	0.9781
K-Nearest Neighbors	n_neighbors: 3, weights: 'distance'	0.9274
Gradient Boosting	learning_rate: 0.2, max_depth: 7, n_estimators: 200	0.9795
AdaBoost	learning_rate: 1, n_estimators: 200	0.8962
Decision Tree	max_depth: 15, min_samples_split: 2	0.9176
CatBoost	depth: 7, iterations: 200, learning_rate: 0.1	0.9777

For evaluation, macro average recall was used as the benchmark for the following reasons:

- Missing a diagnosis (false negative) can be fatal, whereas false positives can be further investigated through additional tests.
- The dataset is imbalanced, so we need a metric not skewed by the dominant class.

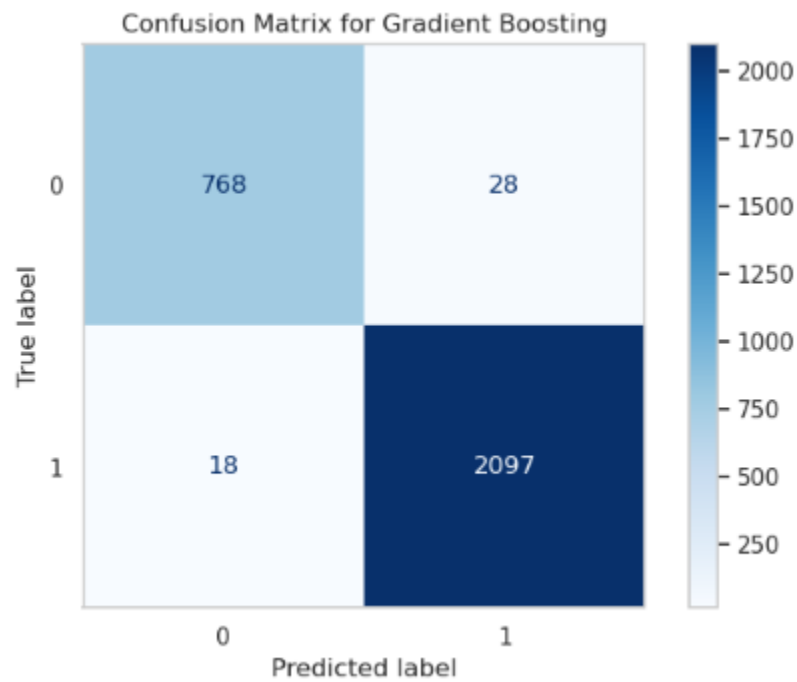
- Macro average recall provides a balanced evaluation across classes.

Best Performing Models: XGBoost and Gradient Boosting

Gradient Boosting - Classification Report

Class	Precision	Recall	F1-score	Support
0.0	0.98	0.96	0.97	796
1.0	0.99	0.99	0.99	2115
Accuracy			0.98	2911
Macro avg	0.98	0.98	0.98	2911
Weighted avg	0.98	0.98	0.98	2911

Confusion Matrix:

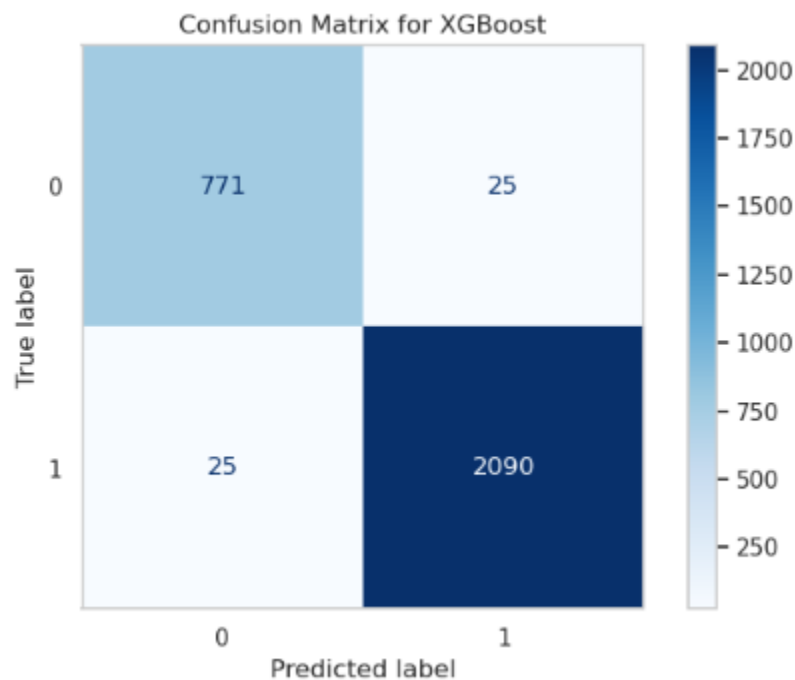


XGBoost - Classification Report

Class	Precision	Recall	F1-score	Support
0.0	0.97	0.97	0.97	796
1.0	0.99	0.99	0.99	2115

Accuracy			0.98	2911
Macro avg	0.98	0.98	0.98	2911
Weighted avg	0.98	0.98	0.98	2911

Confusion Matrix:



Transfer Learning on PTB Diagnostic ECG Dataset

In this study, we employed transfer learning by adapting a pre-trained CNN v2 model—originally developed for the MIT-BIH Arrhythmia Dataset—to the PTB Diagnostic ECG Dataset. Given the PTB dataset's smaller size and binary classification nature (normal vs. abnormal), we made several modifications to the original model:

- **Output Layer Adjustment:** Modified the output layer to accommodate two classes.
- **Data Preprocessing:** Implemented preprocessing steps tailored to the PTB dataset's structure.
- **Model Integration:** Ensured seamless integration of the PTB data into the existing model architecture.

Observations

The same model (**CNN v1**, **CNN v2** and **ResNet1D**) worked for PTB Diagnostic ECG Dataset as well. The **ResNet1D** model stabilizes well after 20 epochs.

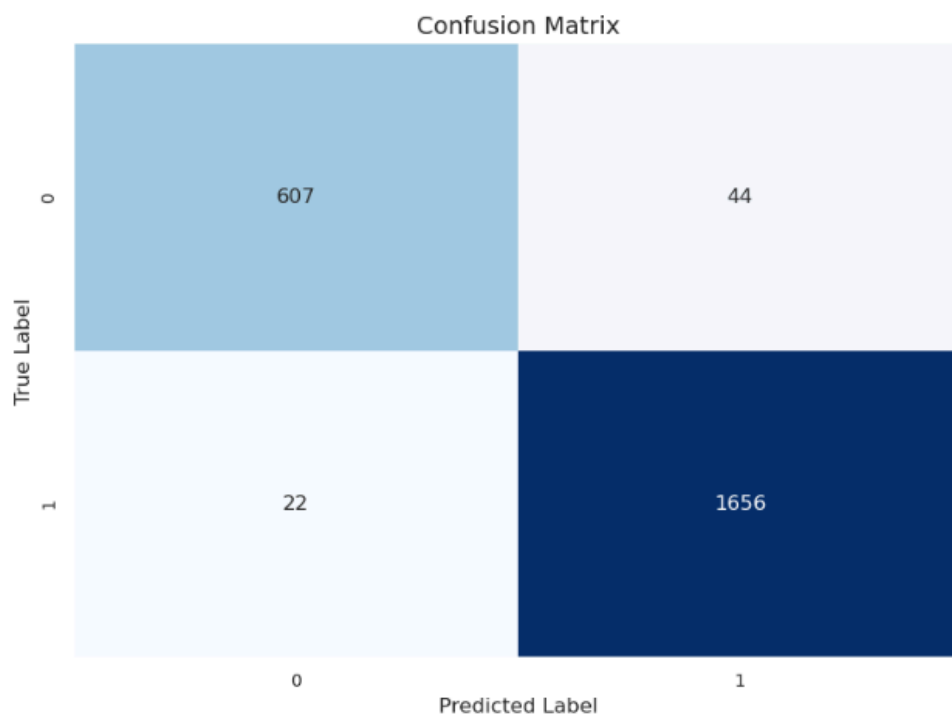
CNN v1



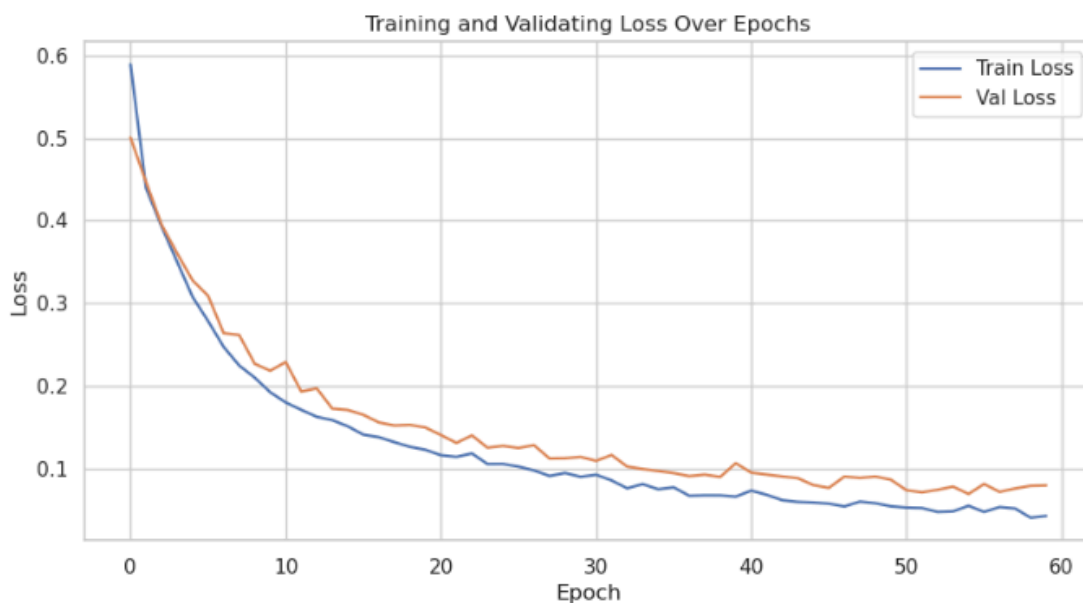
Here is the classification report presented in a structured table format:

Class	Precision	Recall	F1-Score	Support
0	0.965	0.932	0.948	651
1	0.974	0.987	0.980	1678
Accuracy			0.972	2329
Macro Avg	0.970	0.960	0.964	2329
Weighted Avg	0.972	0.972	0.972	2329

Confusion Matrix:



CNN v2

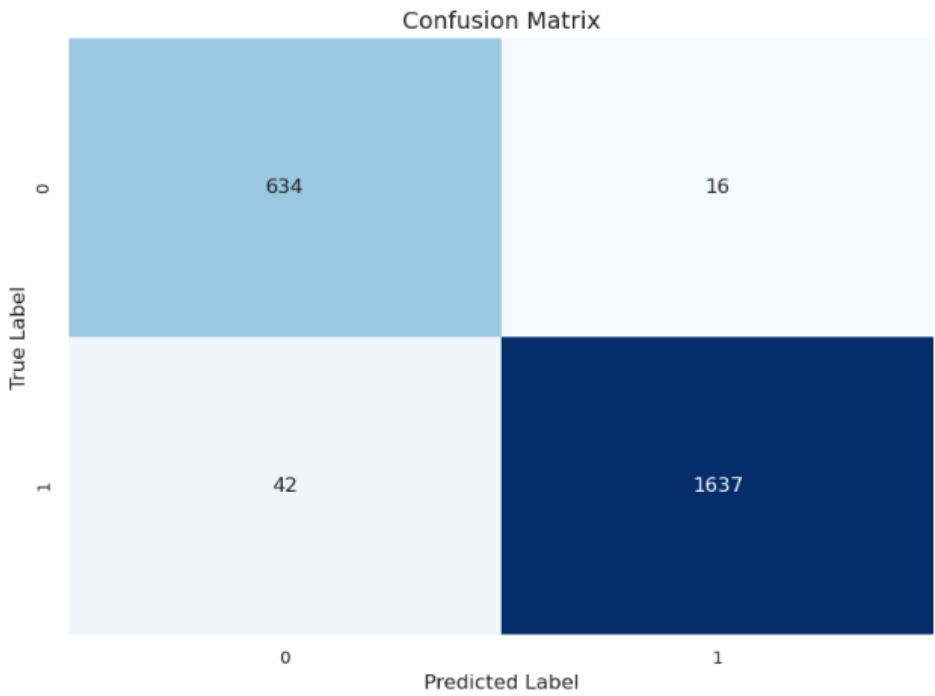


Here is the classification report presented in a structured table format:

Class	Precision	Recall	F1-Score	Support
-------	-----------	--------	----------	---------

0	0.938	0.975	0.956	650
1	0.990	0.975	0.983	1679
Accuracy			0.975	2329
Macro Avg	0.964	0.975	0.969	2329
Weighted Avg	0.976	0.975	0.975	2329

Confusion Matrix:



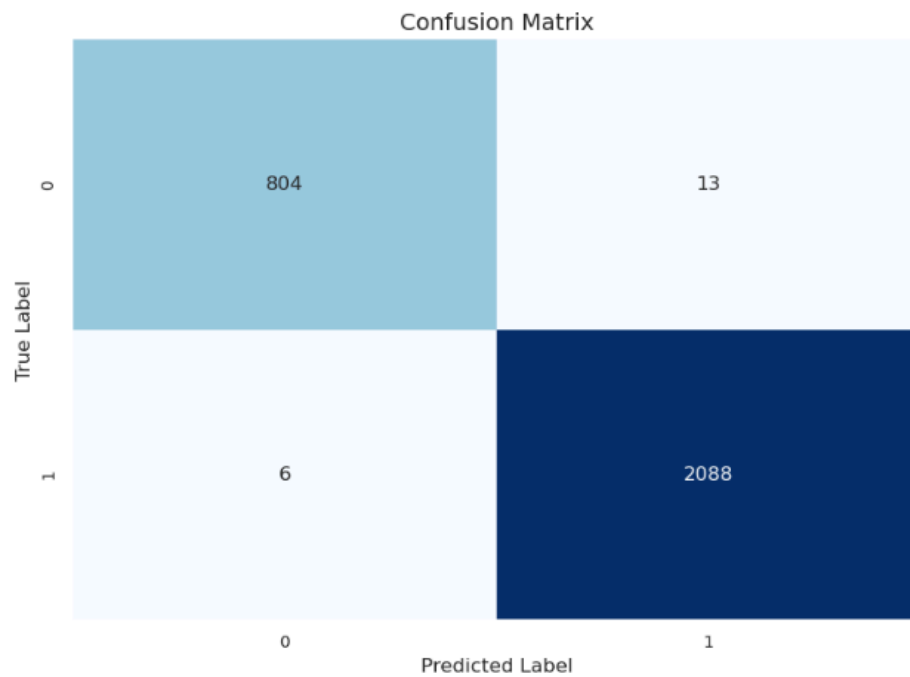
ResNet1D



Here is the classification report presented in a structured table format:

Class	Precision	Recall	F1-Score	Support
0	0.993	0.984	0.988	817
1	0.994	0.997	0.995	2094
Accuracy			0.993	2911
Macro Avg	0.993	0.991	0.992	2911
Weighted Avg	0.993	0.993	0.993	2911

Confusion Matrix:



Model Comparison Using Macro Average Recall

Model	Macro Avg Recall
XGBoost	0.980
Gradient Boosting	0.980
CNN v1	0.960
CNN v2	0.975
ResNet1D	0.991

- **ResNet1D** achieves the highest **macro average recall (0.991)**, indicating the most balanced and effective performance across all classes.
- **CNN v2** follows with a strong recall of **0.975**, showing that deep learning models are outperforming traditional methods in this task.
- **XGBoost** and **Gradient Boosting** both achieve **0.980**, making them solid machine learning baselines.

- **CNN v1**, with **0.960**, has the lowest macro recall, suggesting it is less capable compared to CNN v2 and ResNet1D.

Overall, **deep learning models (CNN v2 and ResNet1D)** show improved generalization and class-wise performance compared to traditional ML approaches.